

MUHAMMAD IRFAN AFIFI BIN MOHD ROMZI

Software Engineer — Full-Stack · IoT Systems · Cloud

+60 13-481 6949 | irfanafifiromzi@gmail.com | Kuala Nerus, Terengganu, Malaysia
[linkedin.com/in/irfanafifiromzi](https://www.linkedin.com/in/irfanafifiromzi) | github.com/irfanafifiromzi | portfolio-irfanafifi.vercel.app



SUMMARY

Software Engineer with 1.5+ years building and shipping production systems across mobile, web and IoT. Delivered an insurance app live on Google Play and the App Store, an RFID-enabled ERP platform, and ESP32-based real-time monitoring systems. Strong in backend and API development with **NestJS** and **Spring Boot**, front-ends in **React**, **Vue.js** and **React Native**, and cloud deployment on **AWS**. Outside work, builds and publishes developer tools on the Microsoft Store, VS Code Marketplace and itch.io.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, PHP, Rust, Dart, SQL

Backend: NestJS, Spring Boot, Node.js, Laravel, REST API design, system integration

Frontend & Mobile: React, Vue.js (Vue 3, Pinia), React Native, Flutter, Electron, Tauri

Cloud & Data: AWS, Huawei Cloud, MySQL (schema design & optimisation)

IoT & Hardware: ESP32, RFID/EPC, environmental & temperature sensors

Tools: Git, SSH, PowerShell, Playwright, Vite, Power BI, Microsoft Office

PROFESSIONAL EXPERIENCE

Evolve Technology Platform Sdn Bhd

Feb 2025 – Present

Software Engineer II

Mar 2026 – Present

- Design and optimise MySQL databases behind ERP and IoT monitoring platforms.
- Deploy and manage production applications on AWS cloud infrastructure.
- Integrate RFID/EPC readers and ESP32-based IoT devices for real-time data collection and monitoring.
- Collaborate with cross-functional teams to deliver end-to-end solutions, from device and backend to dashboard.

Junior Software Engineer

Feb 2025 – Mar 2026

- Developed and maintained enterprise applications across mobile, web and IoT-integrated systems.
- Built scalable backend services and REST APIs with Spring Boot and NestJS.
- Shipped cross-platform mobile features in React Native, released on Google Play and the Apple App Store.
- Created admin dashboards and monitoring systems with Vue.js. Promoted to Software Engineer II after 13 months.

Petroliam Nasional Berhad (PETRONAS)

Jul 2024 – Oct 2024

Intern

- Maintained Power BI dashboards and Excel models for multiple business portfolios, keeping P4R 2024–2029 planning data accurate and current.
- Extracted and consolidated budget transfer data across all 9 pillars and 3 enablers in the department.

KEY PROJECTS — EVOLVE TECHNOLOGY

HAPPI Malaysia Insurance Mobile App

React Native · Vue.js · Spring Boot

- Built mobile app features in React Native and a Vue.js admin panel for internal operations.
- Developed backend REST APIs in Spring Boot (Java); app is live on Google Play Store and Apple App Store.

Zora Pro ERP System

NestJS · TypeScript · React · MySQL · RFID

- Built an ERP covering warehouse, manufacturing and reporting modules on a NestJS backend.
- Integrated RFID/EPC for automated stock tracking, with an admin dashboard for real-time monitoring.

PoleSync Technologies — Smart Environmental Pole

ESP32 · AWS · Vue.js · MySQL

- Smart pole with ESP32 sensors tracking weather, air quality and water quality, doubling as a WiFi access point.
- Turns collected readings into forecasts and feeds a real-time dashboard across all deployed units.

FrostEdge Smart Freezer Monitoring

ESP32 · Node.js · React · MySQL

- Web admin system monitoring freezer temperature and status in real time from ESP32 sensors.

- Backend services process and store sensor data, with automated Telegram and email alerts.

PERSONAL PROJECTS — PUBLISHED

GVReport — Report & Dashboard Designer · [Microsoft Store](#)

Electron · Vue 3 · TypeScript · jsPDF

- Offline-first Windows app for designing pixel-precise paginated reports and multi-page dashboards (charts, KPI tiles, slicers) from Excel/JSON files, live REST endpoints, or MySQL/PostgreSQL queries.
- Built one layout engine that feeds both the on-screen preview and PDF export, so they never disagree; vector charts keep PDFs sharp and small, and batch mode turns one template into many PDFs.
- Wrote a sandboxed expression language (tokenizer, Pratt parser, AST evaluator, no eval) and a read-only SQL allow-list enforced in both the UI and the main process, so a shared project file can never modify a database.
- ~21k lines of TypeScript/Vue, covered by 182 automated pipeline checks and Playwright end-to-end tests; runs on Electron or Tauri (Rust) through a single host abstraction.

GeminiVillain — Developer Command Centre · [itch.io](#), [GitHub](#)

Electron · Vue 3 · node-pty · xterm.js · Rust

- Real PowerShell terminal (ConPTY, multi-tab) that turns saved commands into one-click workflows which stop at the first failed step, so a broken build never reaches production.
- Workflow steps run locally or on remote servers over SSH, carrying the remote exit code back so stop-on-failure works across machines, with a preflight review before anything runs.
- “Is production running my code?” check compares each server’s deployed Git commit with the local branch (up to date, N behind, diverged), plus a status board of read-only health checks.
- One shared Vue frontend runs on either an Electron/node-pty or a Tauri/Rust backend through a common command-and-event contract.

Pixel Pet — VS Code Extension · [VS Code Marketplace](#)

TypeScript · VS Code Extension API · Webview

- Pixel-art aquarium webview panel with hand-drawn animated fish, crab and jellyfish; feed and play commands, click-to-call interaction, keyboard shortcut and user settings.

EDUCATION

Multimedia University (MMU)

Sept 2021 — Oct 2024

Bachelor of Computer Science (Honours), majoring in Software Engineering · CGPA 3.65

Universiti Malaya (UM)

Jun 2020 — Jun 2021

Foundation in Physical Science · CGPA 3.78

LANGUAGES

Malay (Native) · English (Intermediate, written and spoken)